

Advice for 'omics writing

- 1-page summary of your proposed project due on March 3rd
- Today's lab activity is 09_DE_Visualization

Proposing and Publishing 'Omic work

The problem and the challenge:

- 1) Having lots and lots of data can make it hard to summarize
- 2) Many biologists are trained to think about details ... but focusing only on the details is the antithesis of 'omic work
- 3) Focusing only on the details of your own system makes a paper seem narrow (“this study may be more appropriate for a specialty journal”)
- 4) Many of the details are super interesting, but too many points in a manuscript dilutes the main points
 - if you have enough interesting points, maybe separate into 2+ papers

Many genomic proposals and manuscripts turn out to be “descriptive” (or “listy”) and “specialized” without presenting overarching themes or new insights to biology

Publishing 'omic papers

The number 1 pitfall: not laying out a clear rationale for doing 'omics

frontiers in
GENETICS

Death of the genome paper

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One of the drawbacks of genome papers, however, is that they can create a mindset of sequence first, ask questions later. I once attended a Masters thesis defense where the external examiner asked the candidate why he sequenced the chloroplast genome of this particular species and what hypothesis was he trying to test. The student, looking startled, answered, "Because the genome hadn't been sequenced before and we didn't know what it looked like."

The mentor probably had a hypothesis or reason in mind

Publishing 'omic papers

Common pitfalls:

- listing too many details
- having too many points (or not defining the main points)
- not having a clear structure/rationale to guide readers through the details
- not having broad enough conclusions that make the paper interesting to broad readership
- not having anything but details to put in the Discussion

A good paper should have ~3 main points:

- find a way to relate details you're interested into those main points
- many details are interesting in their own right but can't be given equal weight
- many details are interesting to YOU but not people outside your very focused area

Advice: pick an actual person outside your field and imagine you are writing the paper for them

Evolution of transcription factor binding through sequence variations and turnover of binding sites

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Results

Mapping allele-specific binding of 27 transcription factors within an interspecies hybrid

Transcription factors bind a select subset of motif-containing sites within the two genomes

Differential TF binding to the two hybrid alleles correlates with variations in motif sequence, but differences in motif accessibility play a minor role

TF binding to an imperfect motif depends on the genomic context

Sequence variation in the motif predicts TF binding variation

Gains and losses of binding sites are more common than binding-site turnover

Variation in
motif
sequences
plays the
biggest role

Point #3:

Give the reader a framework before you present the details

A common pitfall in writing is to write historically with the result at the end:

We first did this. Then we did something else. We found out one thing. We found out another thing. We did some other analysis and found something else. These results show that we solved cancer.

But a better style that often holds: Pick a good topic sentence that includes the result

We did this and found that the results solve cancer. We first hypothesized this, we led us to this thing. Our results were this, which led us to do something else. Those results hinted at this, so to fully explore the data, we then did something else. Altogether, these results suggest that that answer to cancer is blah (see Discussion)

Many researchers new to scientific writing forget the critical importance of a good topic sentence

Avoid “listing” details and instead create a framework and tell a story that supports the topic sentence

Topic sentence that lists the main conclusion/result

Some guiding language that lays out rationale

Final punchline or interpretation

Next, we sought to understand the major biological processes that were differentially regulated in response to hyposalinity stress. The 60 min time point had the largest number of genes (1480) with the greatest magnitude of expression change, suggesting that this time point best captured the peak of the response. These peak response genes largely returned to their prestress levels of expression by 8 h (Figure 4a). Genes significantly induced at least 2-fold at 60 min were enriched for photosynthetic processes, indicators of oxidative stress, and transcriptional regulation (Figure 4c, Table S7). In contrast, genes significantly repressed at least 2-fold at 60 min were enriched for ribosome biogenesis, transcriptional regulation, translational initiation, protein modification and localization, and ATP metabolism (Figure 4c, Table S7). Repression of genes related to protein synthesis and growth is common in eukaryotic stress responses (Gasch et al., 2000; Mayer et al., 2005) and is probably important for redirecting translational capacity towards induced stress-defense transcripts (Ho et al., 2018; Lee et al., 2011).

General writing advice

1. Make your paper interesting to a broad readership
 - establish what's known (with good references) & remaining questions in the Intro
 - write for someone who's not directly in your field – what would they find interesting?
2. Write using the “Hour Glass Method”
 - start with broad information
 - get more specific as you go
 - relate those specifics back to the big picture
3. The Discussion should not be a second presentation of the Results
 - synergize and synthesize your results with what's already known
 - Is there anything bigger you can say about biology?



The Final Paper

1. Draft of preliminary results due on October 31st, with the final paper being due on December 7th (8-10 pages, single spaced)
2. Introduction: should be no more than 1 - 1 ½ pages.
 - should outline the overall subject, scientific question, how the project will answer your question (e.g., what your predictions are), and why it's important (significance)
 - don't just focus on **experiments**
3. Methods: 1 - 2 pages.
 - should NOT contain more than a paragraph describing the biological data collection
 - should focus on how the data were processed
 - be clear about parameters and details used
 - don't forget references for programs used throughout the course

The Final Paper

3. Results and Discussion (could be separate sections or one section) as if you're writing a real paper.
 - the rest of your paper (4-6 pages)
 - follow the guidelines from this lecture on how to organize (3 key points, don't rehash the Results in the Discussion)
 - include figures and tables when appropriate (as you would for a real paper)
 - make sure the figures add to understanding
 - make sure the figure legends provide a clear description
 - figures should be understandable without having to read any additional text

Quick advice on writing 'omics grant proposals

Of course, many features of a good proposal are independent of the area:

1. Have a compelling and important scientific question (significance)
 - make it clear why your question is important to the broader scientific field
2. Have a great experimental system
 - make it clear why your system is the perfect system to answer your question
3. Have well laid-out aims that stand alone and address the question



genome.gov

National Human Genome Research Institute

National Institutes of Health

Some Dos When Writing Your Application

- Know the requirements of the Funding Opportunity Announcement.
- Propose something significant.
- Discuss why your idea is important.
- Make your research exciting.
- Probe for mechanisms and seek new models.
- Write clearly.
- Be brief with "stuff" that everyone knows.
- Be persistent.
- Write, read, rest X multiple times.

Why Some Research Grant Applications Receive Poor Scores:

- Lack of significance.
- Lack of new or original ideas.
- Absence of an acceptable scientific rationale.
- Lack of experience in the essential methodology.
- Questionable reasoning in experimental approach.
- Uncritical approach or lack of alternative approaches.
- Diffuse, superficial, or unfocused research plan.
- Lack of knowledge of published relevant work.
- Proposal is more work than can be done in reality.
- Lack of grantsmanship.

My top three instant-death triggers in genomics grants

1. Having no rationale for doing 'omics
2. Having no clear hypothesis or prediction about the results
3. Having no clear plan or not enough detail about how the data will be analyzed and how that analysis will lead to important new insights

Laying out a rationale

Experiment: Genome sequencing of some new organism

Rationale: “We predict that there are some genes and genomic features that are shared between our organism and other species, while there are likely other key differences that may contribute to the unique phenotypes and behaviors of our organism”

Experiment: RNA-seq comparing two different tissues in some organism

Rationale: “We predict that there are critical differences in expression between these tissues, and that these differences will underlie the physiological differences in the tissues”

Experiment: Mass-spec identification of proteins that interact with your favorite protein of interest (‘YFP’)

Rationale: “We predict that there are key interacting proteins with YFP that based on functional enrichments and localization will help us better understand YFP’s function”

Clearly lay out your predictions for interpreting the results

“If we see A, we will conclude B. If on the other hand we see C, it may support a model of D”

Make sure you have enough detail about the analysis

1. Know what software tools you will use, and if the method is new, cite why you will use it
 - e.g., xxx reference shows it's the most robust for yyy application
2. Outline the downstream steps after the tool is run
 - e.g., what will you do after you identify differentially expressed genes?
 - *GO enrichment, clustering, etc.*
3. Address false positives and false negatives if proposing a complex experiment
4. Always summarize expected results and the impact of the results
 - What will you get out of the experiment? Is there anything more broadly that these experiments will address?
5. 'Potential pitfalls and alternate approaches' section after each aim
 - *'One possibility is that we will not identify any differentially expressed genes. While this would be unexpected based on xxxx, if this is the case it will still generate important information because it means that the phenotypic differences are due to post-transcriptional mechanisms. In this case, we will focus on yyy'*